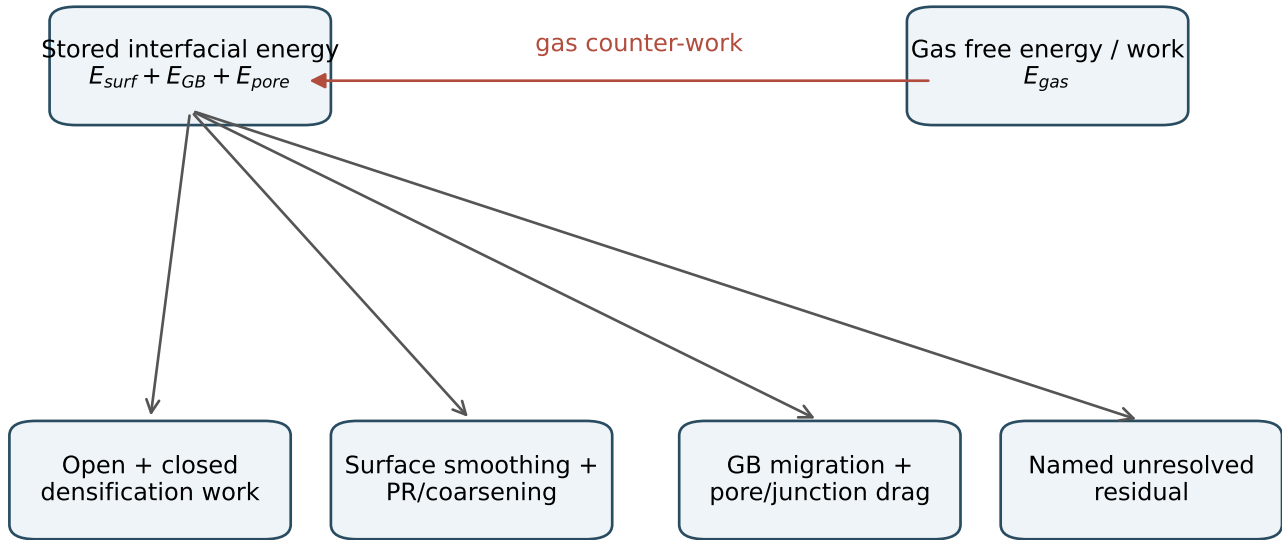


Diagnostic signed energy and dissipation ledger



No channel is silently rescaled; budget violations remain visible.